

Lab 1 warm-up

```
print("Hello World")
```

Hello World

Calculate the volume of a rectangular estuary (length x width x depth)

```
length = 5000    # metres
width = 800      # metres
depth = 3.5      # metres
volume = length * width * depth
print(f"Estuary volume: {volume} m^3")
```

Estuary volume: 14000000.0 m³

Simulate 10 days of exponential population growth

```
population = 100
rate = 0.05
history = [population]

for day in range(10):
    population = population + rate * population
    history.append(round(population, 1))

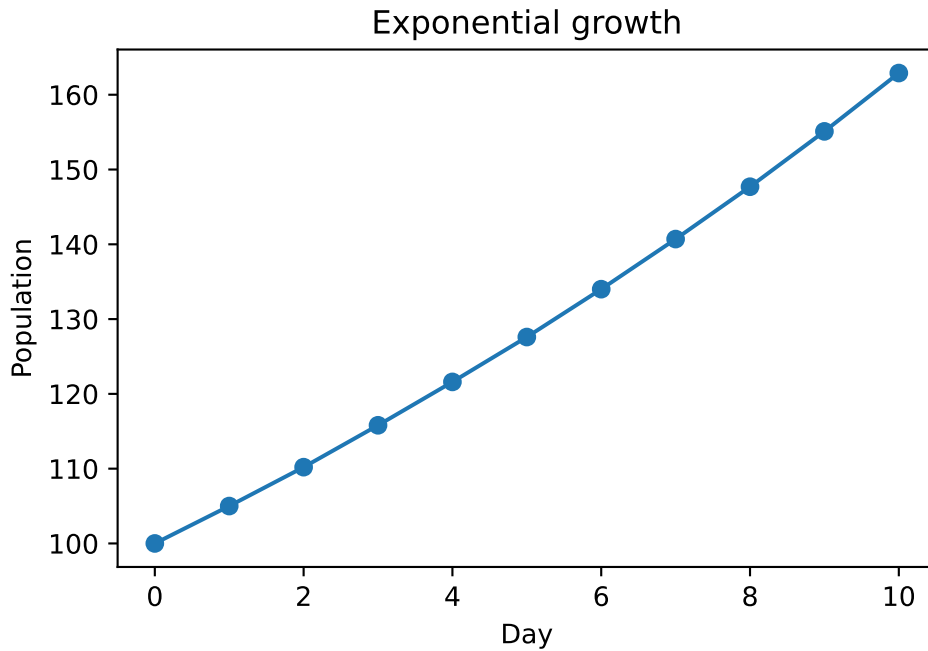
print(history)
```

[100, 105.0, 110.2, 115.8, 121.6, 127.6, 134.0, 140.7, 147.7, 155.1, 162.9]

Write a function that converts temperature from Celsius to Fahrenheit

```
def celsius_to_fahrenheit(c):  
    return c * 9/5 + 32  
  
# Test it  
for temp in [0, 15, 25, 37]:  
    print(f"{temp}°C = {celsius_to_fahrenheit(temp)}°F")  
  
import matplotlib.pyplot as plt  
  
days = list(range(len(history)))  
plt.plot(days, history, "o-")  
plt.xlabel("Day")  
plt.ylabel("Population")  
plt.title("Exponential growth")  
plt.show()
```

0°C = 32.0°F
15°C = 59.0°F
25°C = 77.0°F
37°C = 98.6°F



```
import matplotlib.pyplot as plt

concentration = 100.0
decay_rate = 0.01
dt = 1.0
history = [concentration]

for day in range(60):
    concentration = concentration - decay_rate * concentration * dt
    history.append(concentration)

plt.plot(history)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake")
plt.show()
```

